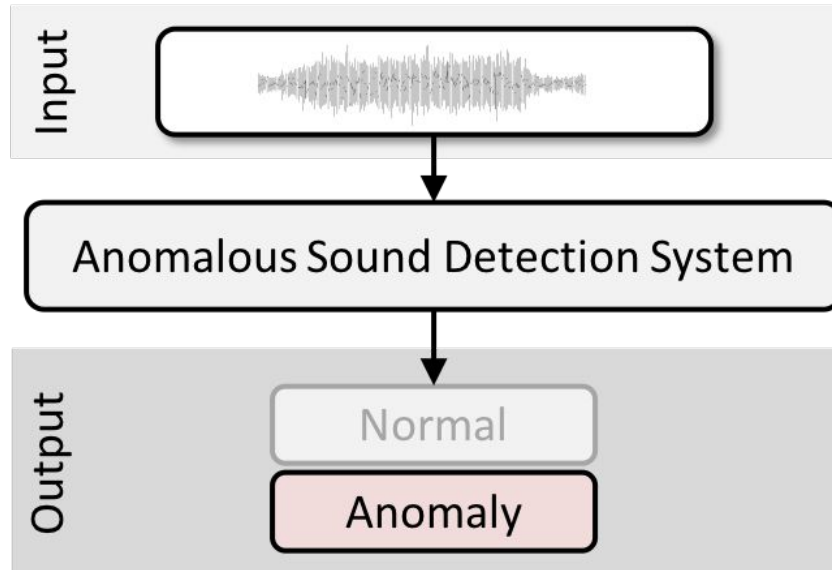


# Detección de Anomalías Acústicas en Maquinaria Industrial

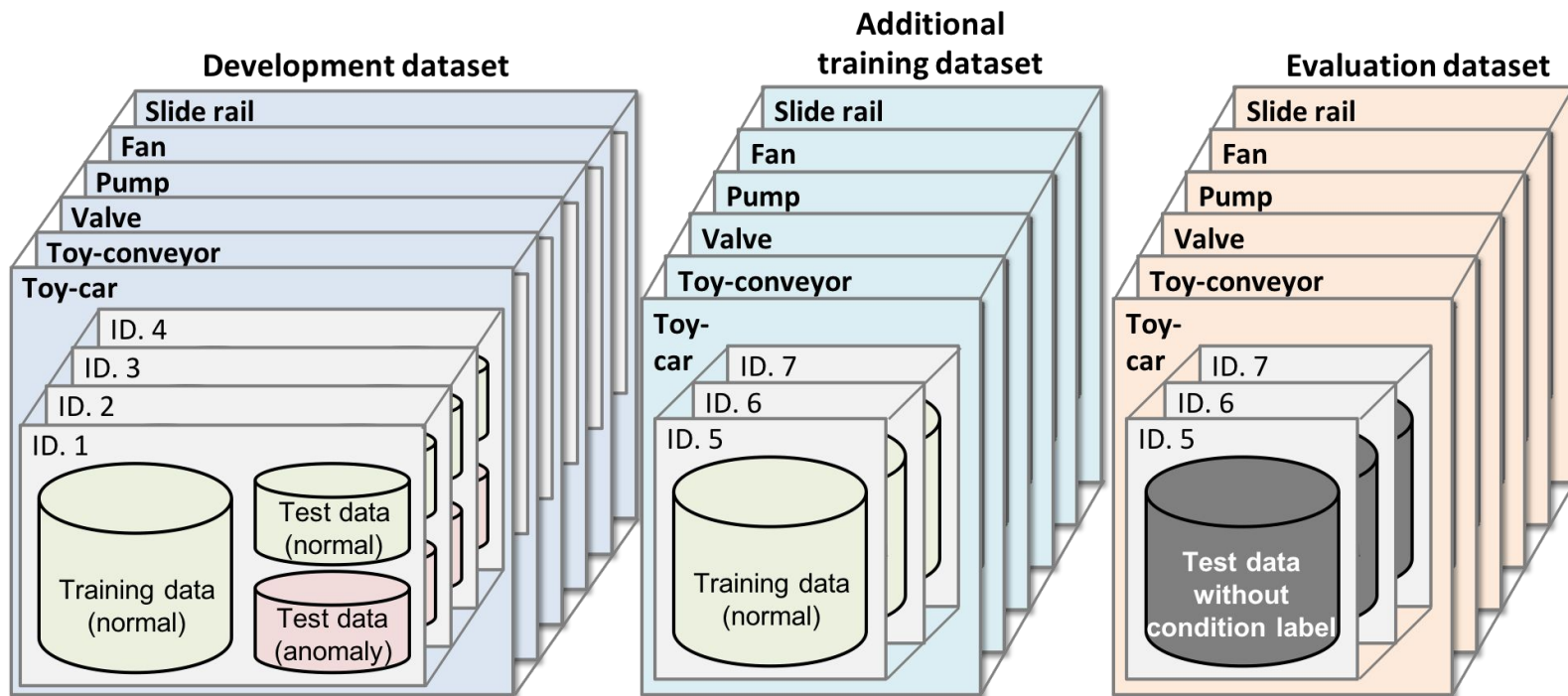
Arley Andrés Villamizar Molina

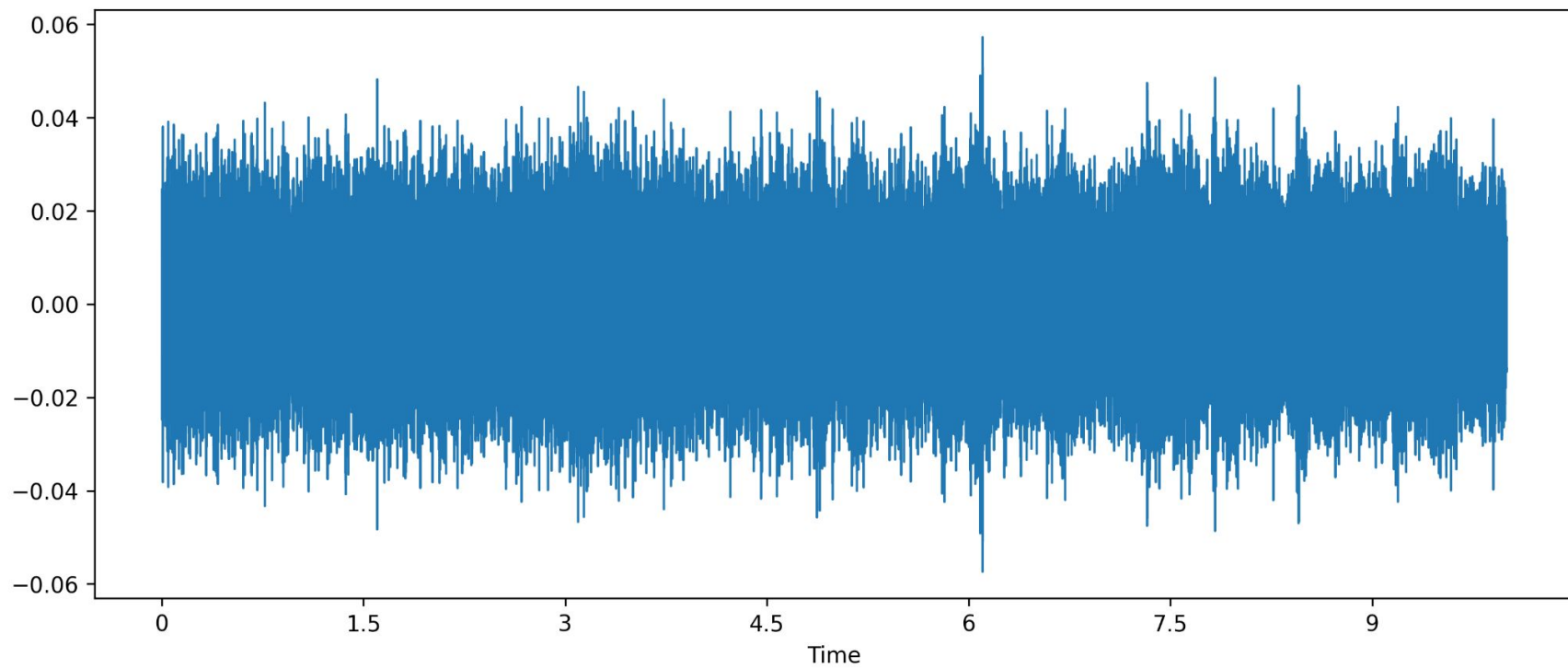
# Alcance

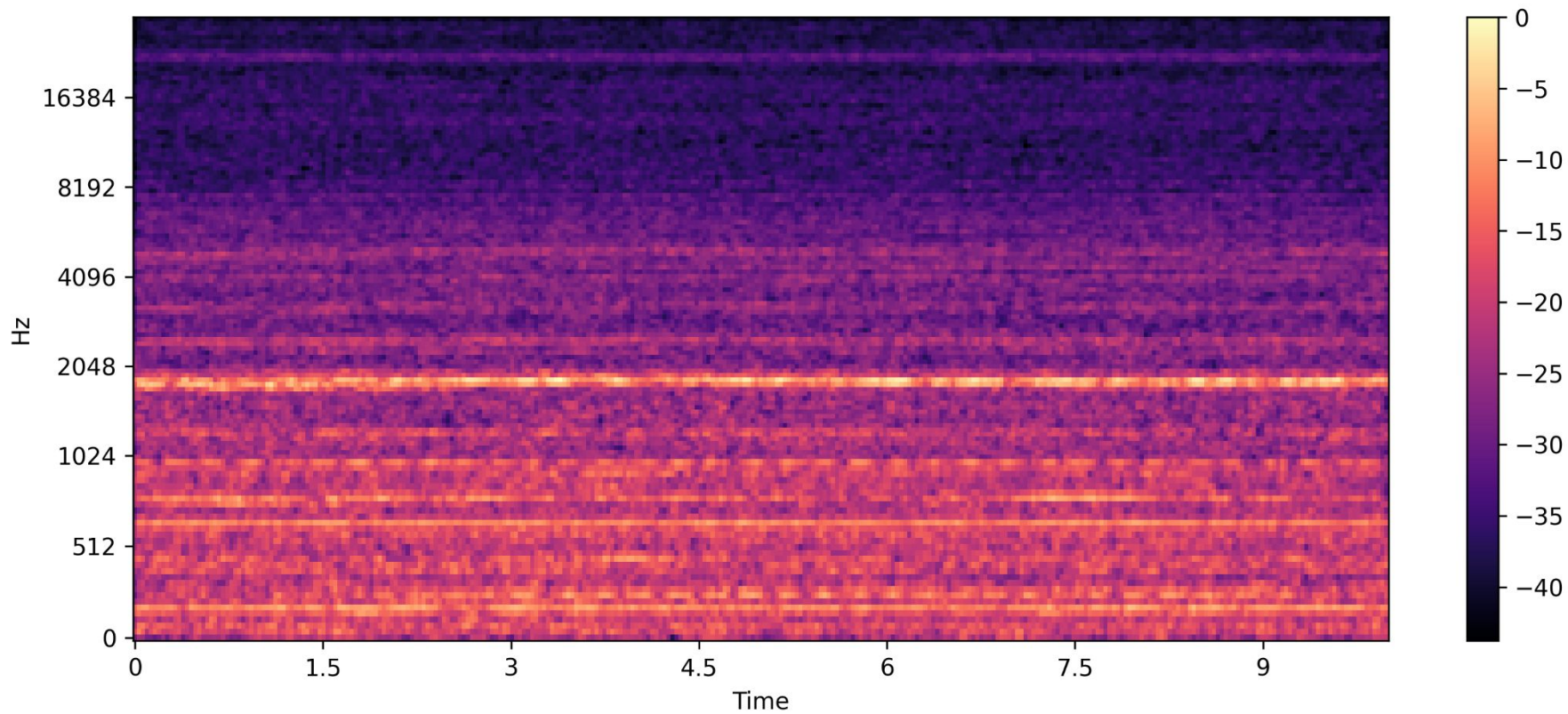


Detectar anomalías  
acústicas en maquinaria  
industrial haciendo uso  
de autoencoders

# Dataset



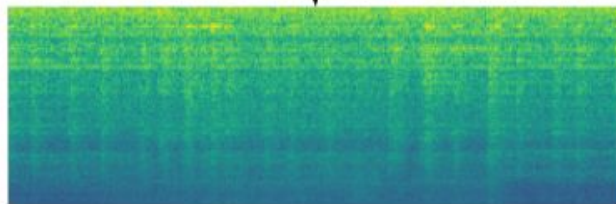




Audio Signal



Log-mel energies spectrogram

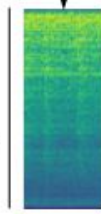


Buffer of 1 second

0.9 seconds overlap

Buffer of 1 second

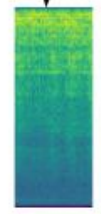
128 MFECs



32 frames

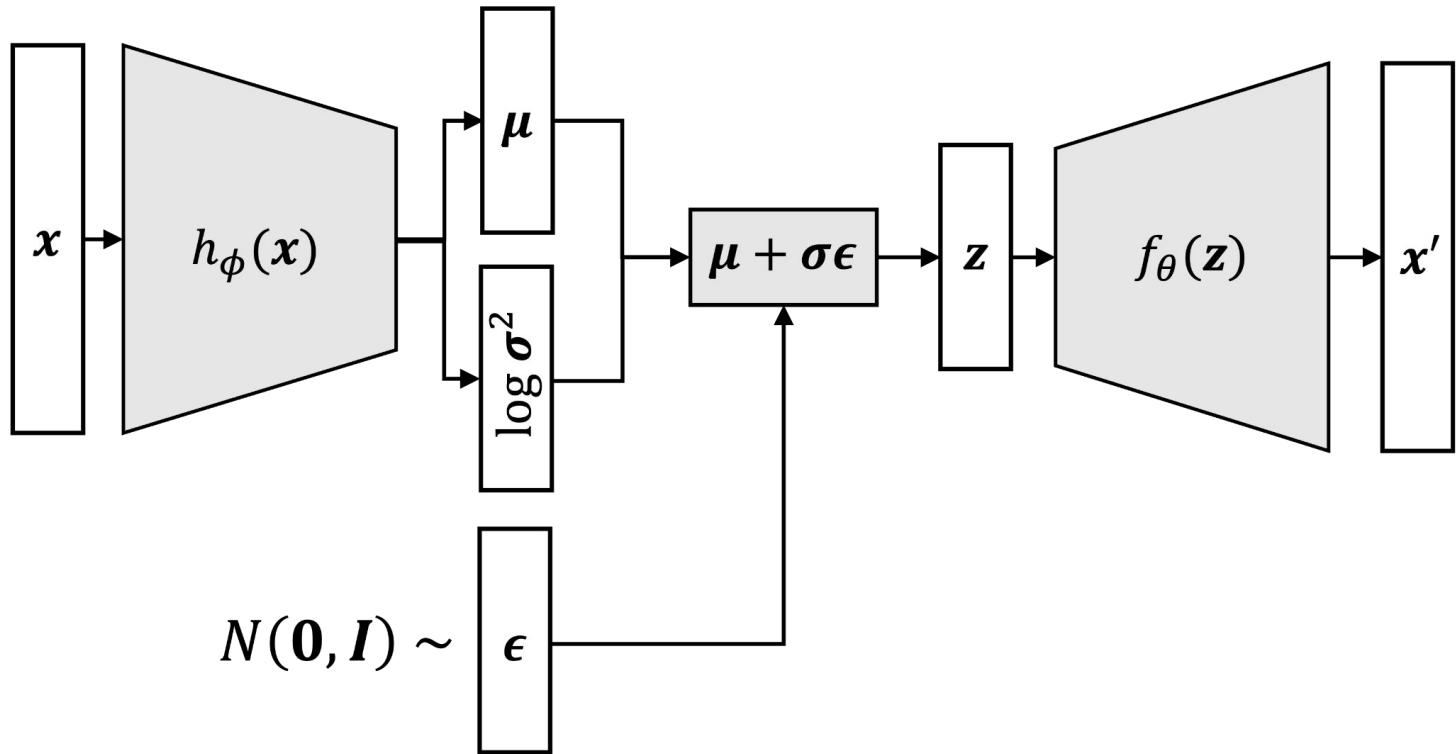
N data segments

128 MFECs



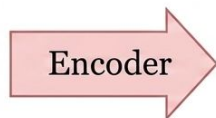
32 frames

# Variational Autoencoder

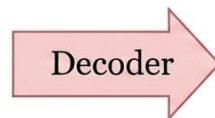




Input image


$$\begin{bmatrix} -0.23343 \\ 0.57239 \\ 0.12739 \\ -0.77489 \\ 0.38380 \end{bmatrix}$$

Encoding

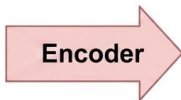


Reconstruction





Input image



Encoding in 'latent space'

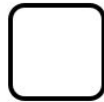
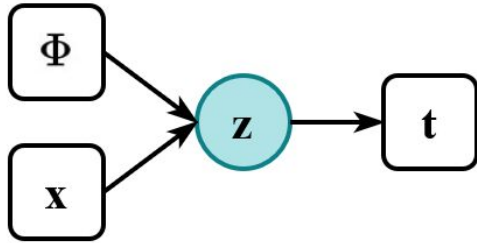
$$\begin{bmatrix} 0.349 \\ 0.929 \\ -0.299 \\ 1.690 \\ 0.380 \end{bmatrix} \quad \begin{bmatrix} 4.12 \\ 2.33 \\ 1.12 \\ 3.24 \\ 2.05 \end{bmatrix}$$

Means  
 $\mu$

Stddev  
 $\sigma$

$$\begin{bmatrix} X(0.349, 4.12^2) \\ X(0.929, 2.33^2) \\ X(-0.299, 1.12^2) \\ X(1.690, 3.24^2) \\ X(0.380, 2.05^2) \end{bmatrix} \xrightarrow{\text{Sample}} \begin{bmatrix} 0.602 \\ 2.331 \\ -0.001 \\ 6.349 \\ 2.187 \end{bmatrix}$$

### Original Formulation

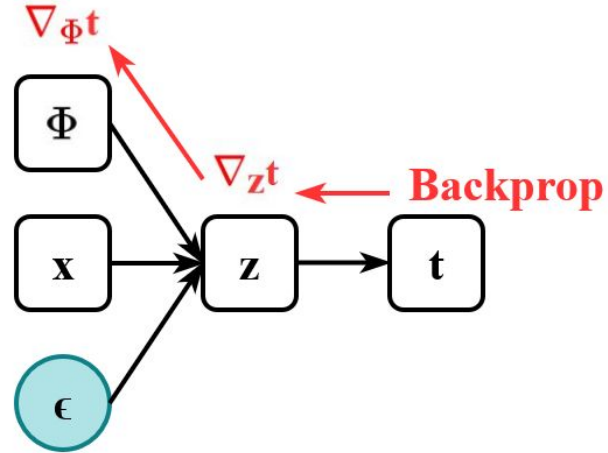


Deterministic  
Node



Random  
Node

### Reparameterized Formulation



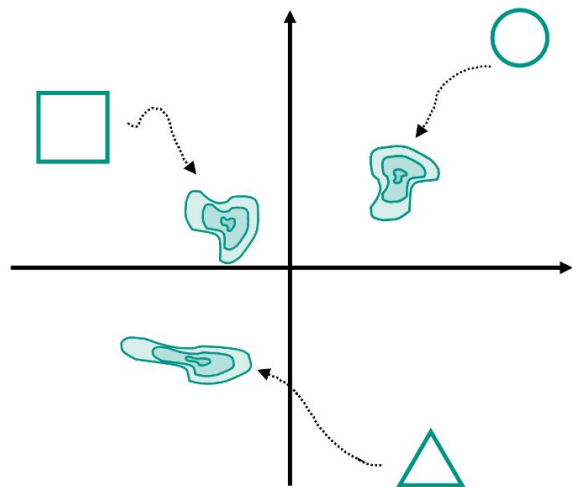
Feedforward



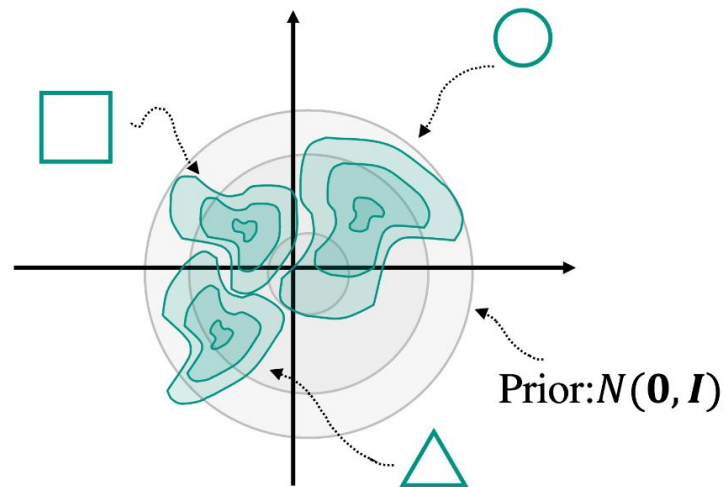
Differentiation

$$\text{loss}_{\text{VAE}}(\phi, \theta) = - \sum_{i=1}^n E_{\mathbf{z}_i \sim q_{\phi}(\mathbf{z}_i | \mathbf{x}_i)} [\log p_{\theta}(\mathbf{x}_i | \mathbf{z}_i)] - KL(q_{\phi}(\mathbf{z}_i | \mathbf{x}_i) || p(\mathbf{z}_i))$$

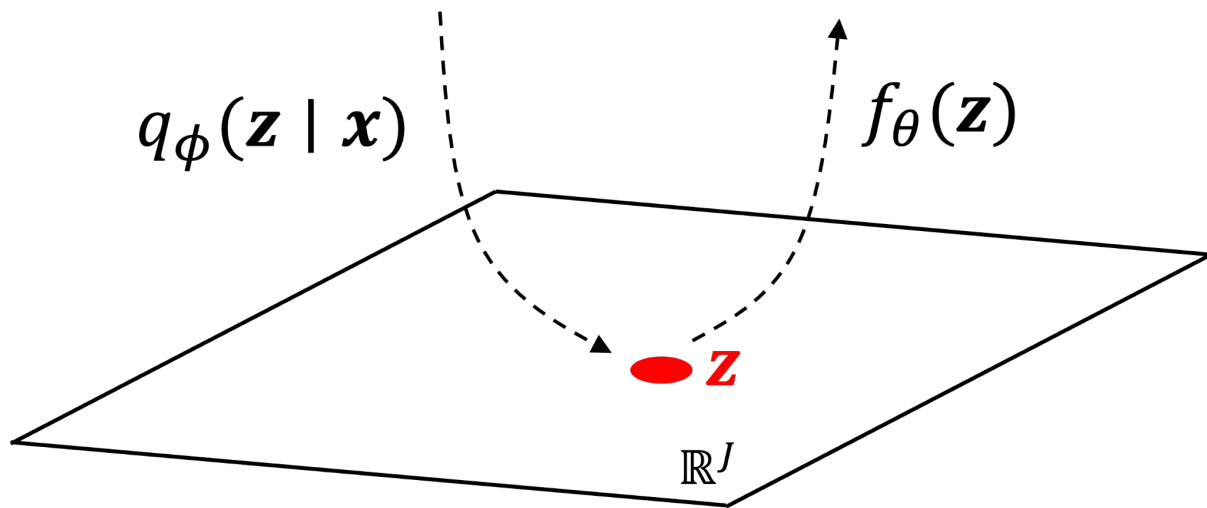
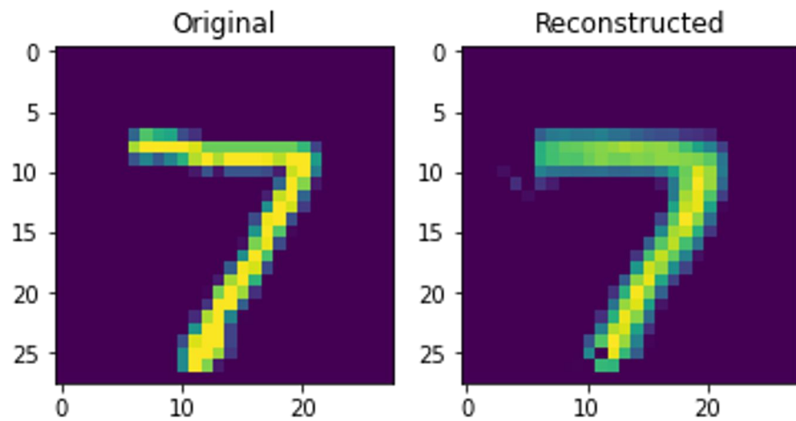
$$KL(q_{\phi}(\mathbf{z} | \mathbf{x}) || p(\mathbf{z})) = -\frac{1}{2} \sum_{j=1}^J \left( 1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2 \right)$$

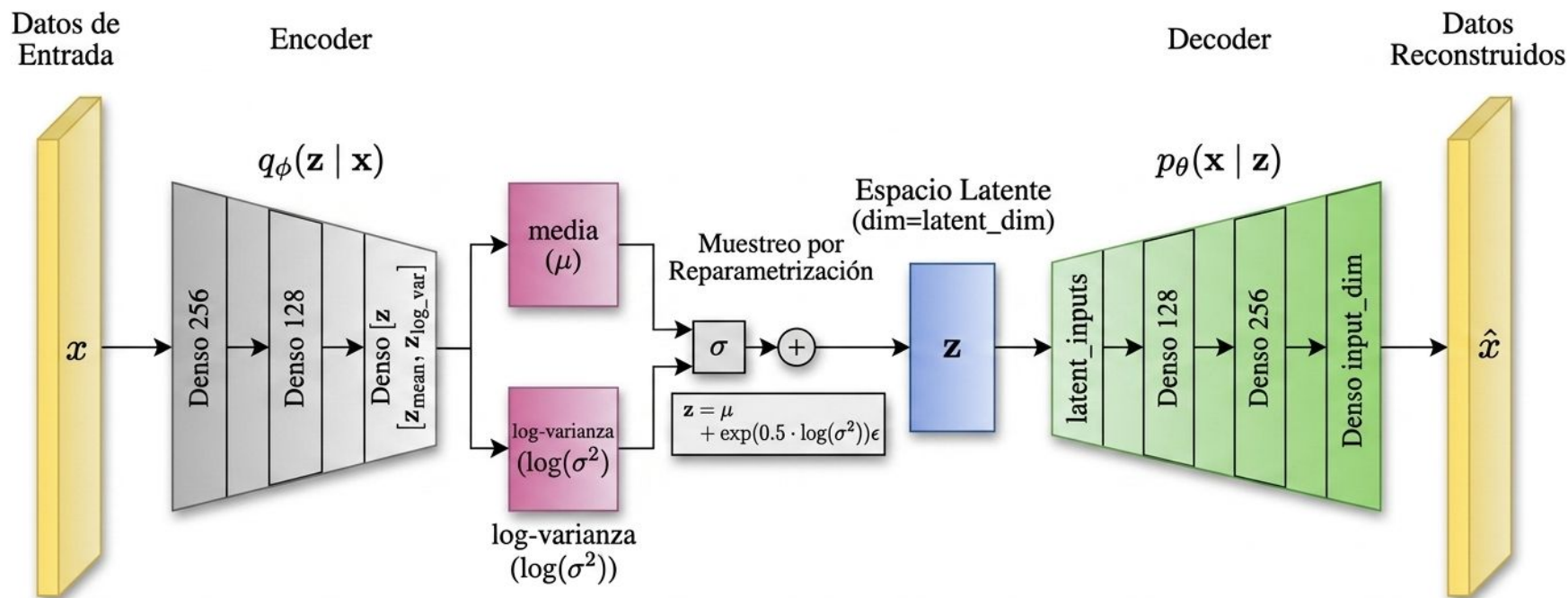


AE latent space



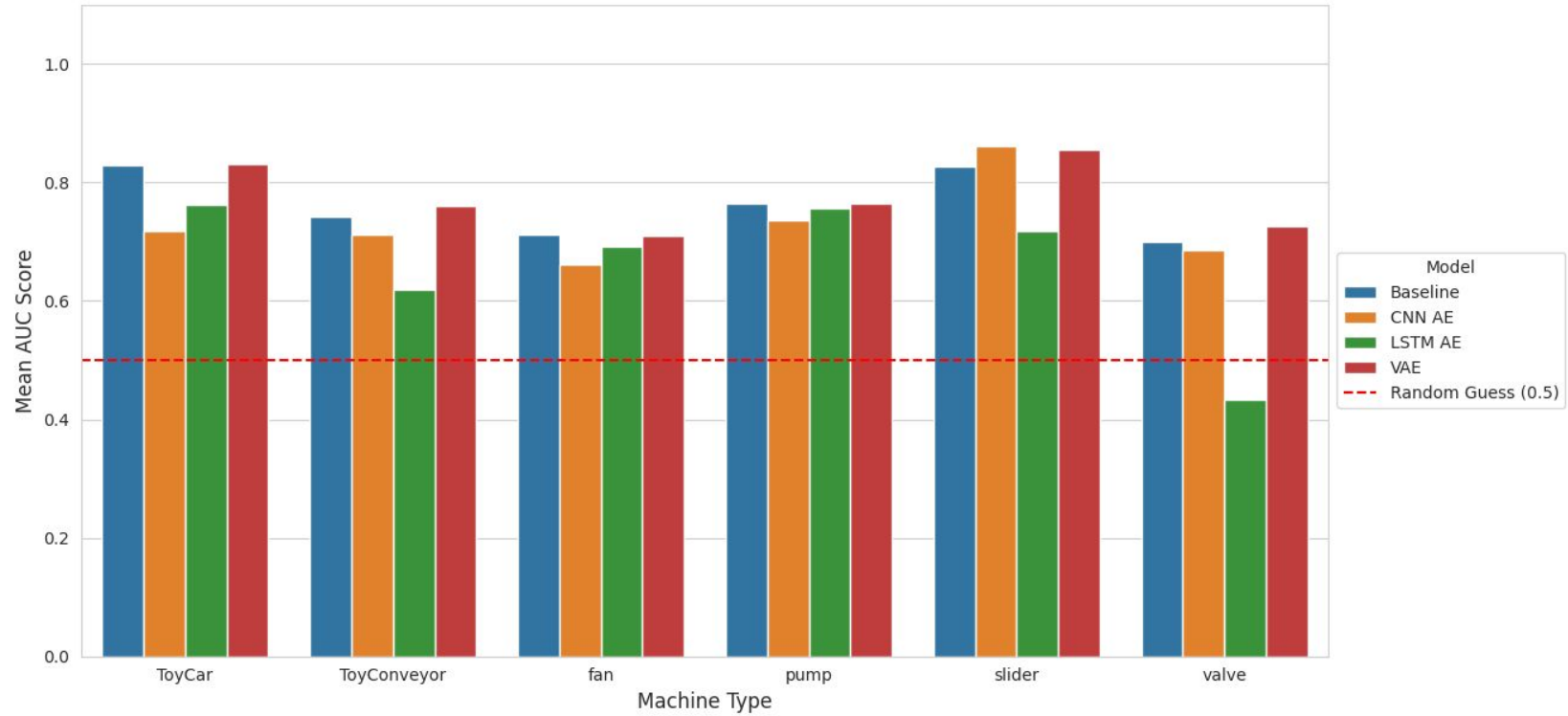
VAE latent space





$$\text{Pérdida Total} = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z}|\mathbf{x})} [\log p(\mathbf{x}|\mathbf{z})] - 0.5 \cdot \sum (1 + \log(\sigma^2) - \mu^2 - \exp(\log(\sigma^2))) + \text{Divergencia KL}$$

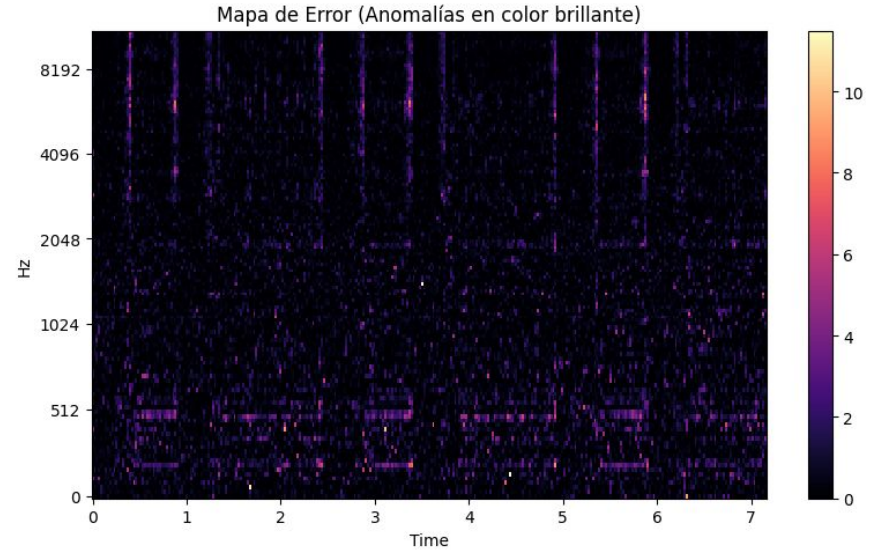
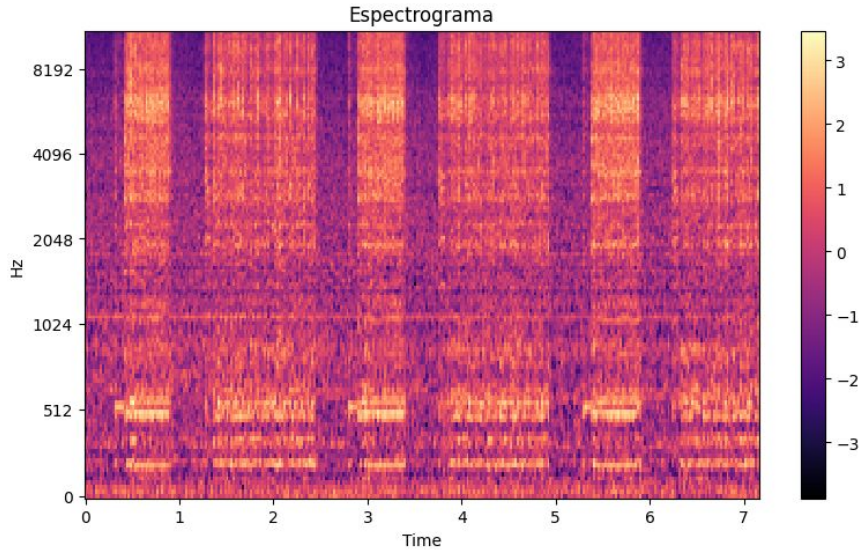
Comparación de Modelos por Tipo de Máquina (Mean AUC)



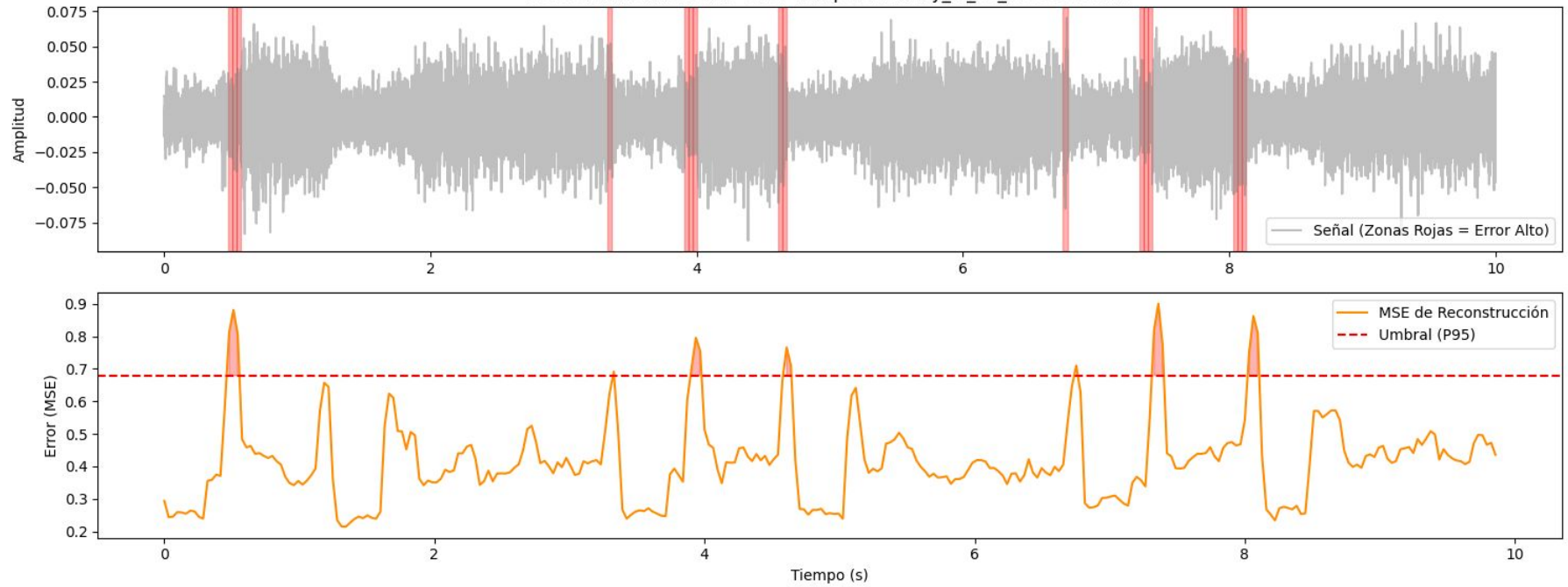
| <b>Modelo</b> | <b>Promedio AUC</b> |
|---------------|---------------------|
| AE            | 0.763728            |
| CNN AE        | 0.730149            |
| LSTM AE       | 0.666109            |
| <b>VAE</b>    | <b>0.775723</b>     |



# Mapa de Errores



Detección de Anomalías en el Tiempo: anomaly\_id\_00\_00000000.wav



# Notebook